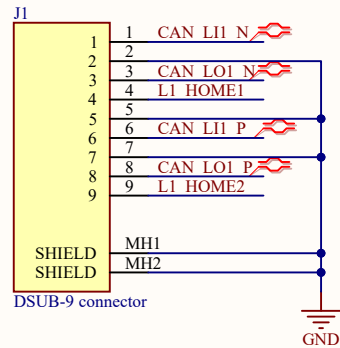
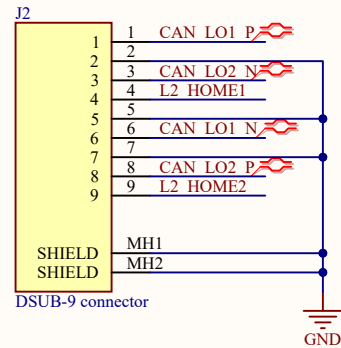
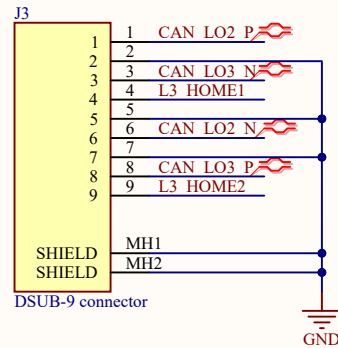
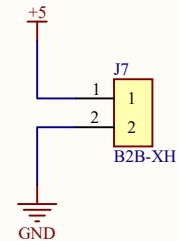
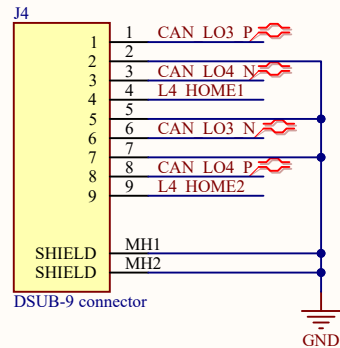
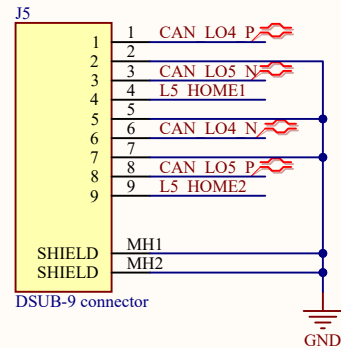
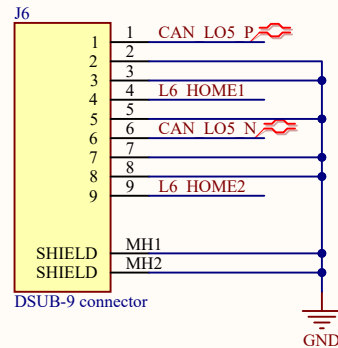
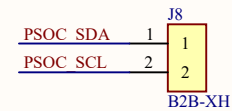
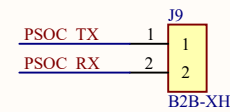


DSUB9 LEG 1**DSUB9 LEG 2****DSUB9 LEG 3****5V SUPPLY CONNECTOR****DSUB9 LEG 4****DSUB9 LEG 5****DSUB9 LEG 6****PSOC5 I2C****PSOC5 UART**

Title: Connectors			
Project: C.A.N.I.S.			
State: Production	Organization: Embedded Dynamics		
Id: ED4		Size: A4	Rev: 3
Author: P. Huijsen	Date: 30-03-26	Sheet: 1 of 2	

PSoC MAINBOARD

The diagram illustrates the PSoC MAINBOARD, centered around the CY8CKIT-059 microcontroller. The microcontroller is shown as a large yellow rectangle with various pins and internal blocks labeled. Key components and connections include:

- Power Supply:** A +5V supply is connected to the VDDIO and VDD pins (J1.26, J2.1). A 100nF capacitor (C5) is connected to the +5V supply and GND.
- Reset:** The RESET pin (J2.3) is connected to the microcontroller's RESET pin.
- GPIO Pins:** Numerous GPIO pins are connected to headers J1.1 through J1.24 and J2.1 through J2.26. These include pins for P0.0/GPIO, P0.1/GPIO, P0.2/GPIO/BYPASSCAP, P0.3/GPIO/BYPASSCAP, P0.4/GPIO/BYPASSCAP, P0.5/GPIO, P0.6/GPIO, P0.7/GPIO, P1.0/GPIO, P1.1/GPIO, P1.2/GPIO, P1.3/GPIO, P1.4/GPIO, P1.5/GPIO, P1.6/GPIO, P1.7/GPIO, P2.0/GPIO, P2.1/GPIO/LED, P2.2/GPIO/SW, P2.3/GPIO, P2.4/GPIO, P2.5/GPIO, P2.6/GPIO, P2.7/GPIO, P3.5/GPIO, P3.6/GPIO, P3.2/GPIO/BYPASSCAP, P3.4/GPIO, P3.3/GPIO, P3.0/GPIO, P3.1/GPIO, P3.7/GPIO, P12.0/GPIO/I2C_SCL, P12.1/GPIO/I2C_SDA, P12.2/GPIO, P12.3/GPIO, P12.4/GPIO, P12.5/GPIO, P12.6/GPIO/UART_RX, P12.7/GPIO/UART_TX, P15.0/GPIO, P15.1/GPIO, P15.2/GPIO/XTAL_OUT, P15.3/GPIO/XTAL_IN, P15.4/GPIO/CMOD, and P15.5/GPIO.
- UART:** The PSOC_RX (J1.10) and PSOC_TX (J1.9) pins are connected to the microcontroller's UART pins.
- I2C:** The PSOC_SCL (J1.16) and PSOC_SDA (J1.15) pins are connected to the microcontroller's I2C pins.
- Home1 and Home2 Pins:** L1 HOME1 (J2.10), L1 HOME2 (J2.11), L2 HOME1 (J2.21), L2 HOME2 (J2.22), L3 HOME2 (J2.25), L3 HOME1 (J2.24), L4 HOME1 (J1.14), L4 HOME2 (J1.13), L5 HOME1 (J1.12), and L5 HOME2 (J1.11) are connected to the microcontroller's pins.
- Canal RX and TX:** The CAN_RX (J2.5) and CAN_TX (J2.4) pins are connected to the microcontroller's pins.
- XTAL:** The XTAL_N (J2.17) and XTAL_P (J2.12) pins are connected to the microcontroller's XTAL pins. A 20MHz crystal (Y1) is connected between these pins. A 18pF capacitor (C3) is connected to XTAL_P and GND, and another 18pF capacitor (C4) is connected to XTAL_N and GND.
- Grounding:** GND pins (GND_J1, GND_J2, GND_J2) are connected to the microcontroller's ground pins.

The microcontroller is labeled CY8CKIT-059 at the bottom left.

CAN TRANSCEIVER

The diagram illustrates a CAN Transceiver circuit using a TJA1040 IC (U1). The IC is a yellow square component with pins labeled 3, 4, 5, 6, 7, and 8. Pin 3 is connected to VCC (+5V), and pin 4 is connected to GND. Pin 5 is connected to GND. Pin 6 is connected to CAN LI1_N, and pin 7 is connected to CAN LI1_P. Pin 8 is connected to STB. The IC has four inputs: CAN TX (pin 1), CAN RX (pin 2), TXD (pin 3), and RXD (pin 4). The IC has four outputs: CANL (pin 5), SPLIT (pin 6), CANH (pin 7), and CANL (pin 8). The circuit includes two 100nF capacitors (C1 and C2) connected to the CAN LI1_N and CAN LI1_P lines. Two 60Ω resistors (R1 and R2) are connected to the CAN LI1_N and CAN LI1_P lines. The CAN LI1_N line is also connected to a red wire, and the CAN LI1_P line is connected to a red wire. The CAN TX and CAN RX lines are connected to a red wire. The TXD and RXD lines are connected to a red wire. The STB line is connected to a red wire. The CANL and SPLIT lines are connected to a red wire. The CANH line is connected to a red wire.

Title: Microcontroller			
Project: C.A.N.I.S.			
State: Production	Organization: Embedded Dynamics		
Id: ED5	Size: A4	Rev: 3	
Author: P. Huijser	Date: 30-03-26	Sheet: 2	of 2

Comment	Description	Designator	Footprint	LibRef	Quantity
100nF	SMD Capacitor	C1, C2, C5	CAP_0805	CAP_0805	3
18pF	SMD Capacitor	C3, C4	CAP_0805	CAP_0805	2
DSUB-9 connector	THT DSUB-9 connector	J1, J2, J3, J4, J5, J6	ADF09AKGT2	DSUB-9 connector	6
B2B-XH	Connector	J7, J8, J9	B2BXH	B2B-XH	3
60Ω	SMD Resistor	R1, R2	RES_0805	RES_0805	2
TJA1040	1/1 Transceiver Half CANbus 8-SO	U1	SOIC127P600X175-8N	TJA1040	1
CY8CKIT-059	CY8C58LP PSOC5LP Embedded Evaluation Board	U2	MODULE_CY8CKIT-059	CY8CKIT-059	1
20MHz	16 MHz $\pm$ 30ppm Crystal 18pF 50 Ohms HC-49/US	Y1	XTAL_HC49P488W43L1150T466H350	HC-49/U-S16000000ABJB	1